



HL-PT034B-N(WH)

Phototransistor

HL-PT034B-N(WH) is a high speed and high sensitive NPN silicon phototransistor moulded in a 5mm (T1/34) black through hole plastic package. The device is sensitive to visible and infrared radiation LED.

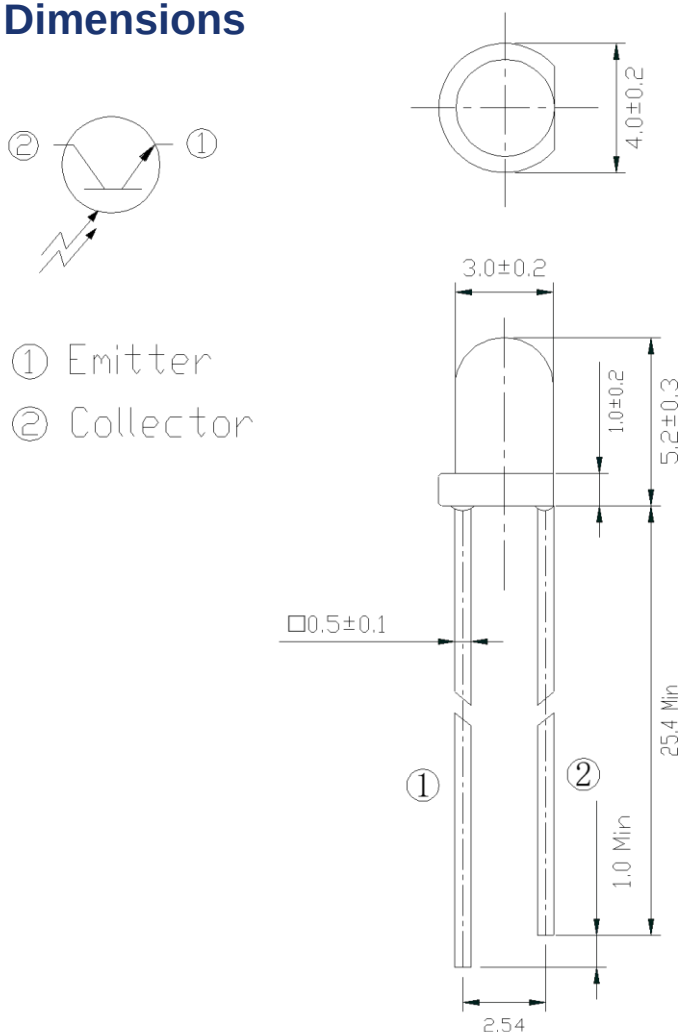
Features

- High photo sensitivity
- Fast response time
- Pb Free, RoHS and REACH compliant version
- Offered as a suitable alternative to Rohm RPT38PB3F

Applications

- Infrared applied system
- Infrared detection

Package Dimensions



*All dimensions are in millimeters.

*Tolerance is $\pm 0.3\text{mm}$ unless otherwise specified

Absolute Maximum Ratings

Parameter ($T_a=25^\circ\text{C}$)	Symbol	Ratings	Unit
Collector-Emitter Voltage	V_{CEO}	30	V
Emitter-Collector-Voltage	V_{ECO}	5	V
Collector Current	I_c	20	mA
Collector Power Dissipation	P_D	75	mW
Operating Temperature	T_{opr}	-25~+85	$^\circ\text{C}$
Storage Temperature	T_{stg}	-40~+100	$^\circ\text{C}$
Lead Soldering Temperature *1 (2mm form body for 5 seconds)	T_{sol}	260	$^\circ\text{C}$

*1. Soldering time $\leq 5\text{seconds}$



Electro-Optical Characteristics

Parameter (Ta=25°C)	Symbol	Condition	Min.	Typ.	Max.	Units
Rang of Spectral Bandwidth	$\lambda_{0.1}$	---	700	---	1100	nm
Wavelength of Peak Sensitivity	λ_P	---	---	800	---	nm
Collector-Emitter Breakdown Voltage	BV_{CEO}	$I_C=100\mu A$ $E_e=0mW/cm^2$	30	---	---	V
Emitter-Collector Breakdown Voltage	BV_{ECO}	$I_E=100\mu A$ $E_e=0mW/cm^2$	5	---	---	V
On State Collector Current	$I_{C(ON)}$	$V_{CE}=5V$ $\lambda_P=940nm$ $E_e=1mW/cm^2$	0.70	---	---	mA
Collector Dark Current	I_{CEO}	$V_{CE}=20V$ $E_e=0mW/cm^2$	---	---	100	nA
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$	$I_C=1mA$ $\lambda_P=940nm$ $E_e=1mW/cm^2$	---	---	0.4	V
Rise Time	t_r	$V_{CE}=5V$ $I_C=1mA$	---	15	---	μS
Fall Time	t_f	$R_L=1000\Omega$	---	15	---	

Bin Rank

BIN	Symbol	Condition	Min.	Max.	Unit
G	$I_{C(ON)}$	$V_{CE}=5V$ $\lambda_P=880nm$ $E_e=1mW/cm^2$	0.70	1.90	mA
H			1.14	2.6	
J			1.77	3.61	
K			2.67	5.07	
L			4.18	7.07	

Notes: 1. Tolerance of On State Collector Current $\pm 10\%$.

Typical Electro-Optical Characteristics Curves for PT

Fig. 1 Collector Power Dissipation vs. Ambient Temperature

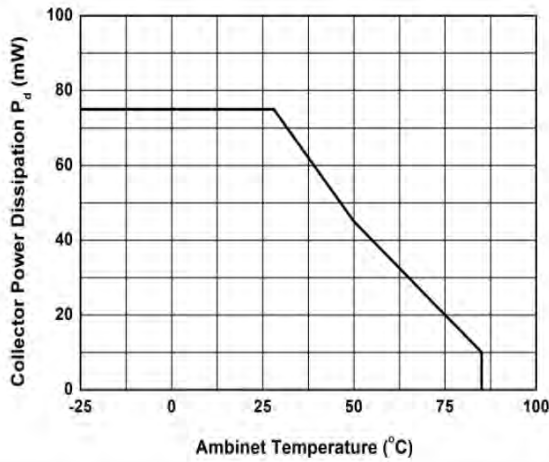


Fig. 2 Collector Dark Current vs. Ambient Temperature

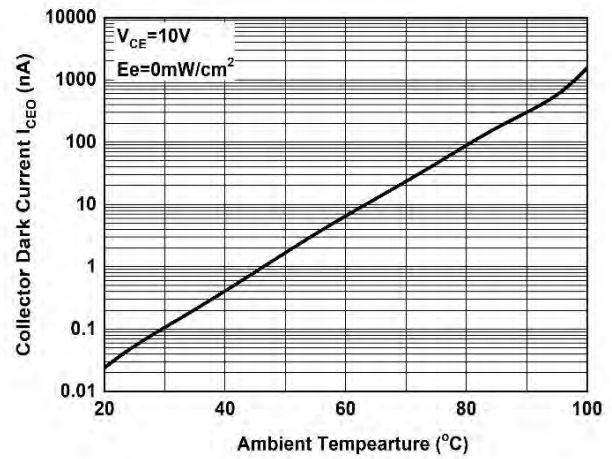
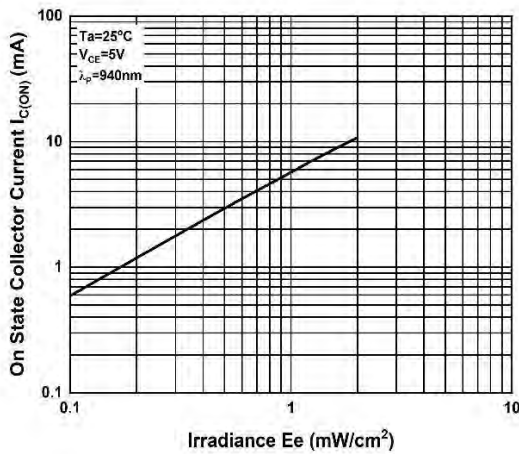
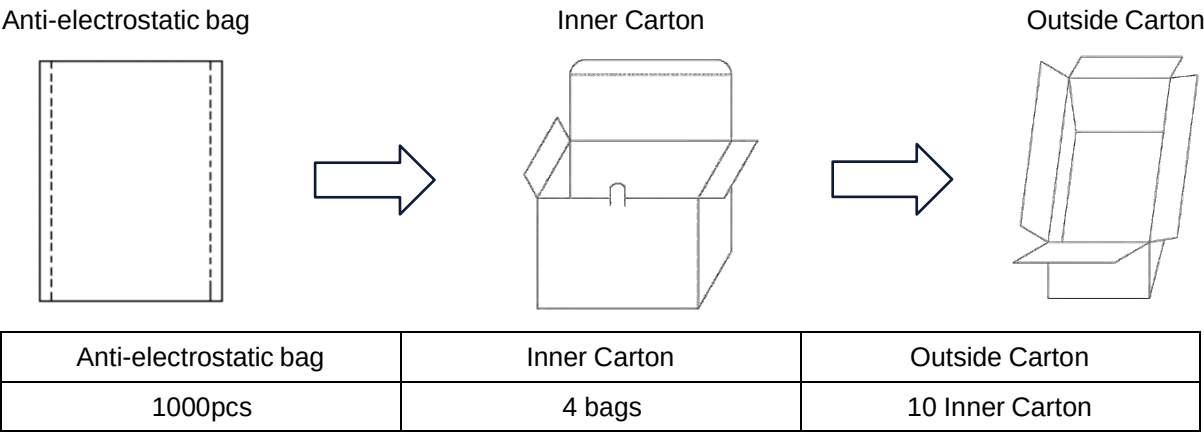


Fig. 3 Collector Current vs. Irradiance E_e





Packing Specification



Weight: 170 g

Product Label



CPN: Customer's Product Number
P/N: Product Number
QTY: Number of Component Per Reel
LOT NO: Lot Number
D/C: DATE



Soldering

Over-current-proof

Current-limiting resistors must be used to prevent the LED from failure due to excessive current.

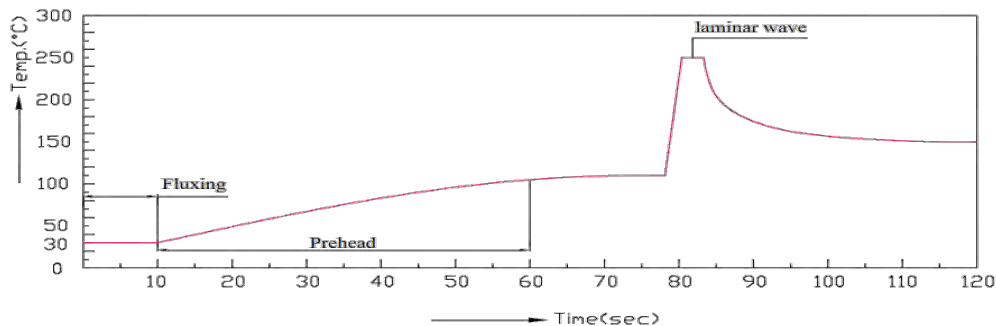
Soldering Condition

1. Careful attention should be paid during soldering. When soldering, leave more than 3mm from solder joint to epoxy bulb, and soldering beyond the base of the tie bar is recommended.

2. Recommended soldering conditions:

Hand Soldering		DIP Soldering	
Temp. at tip of iron	300°C Max. (30W Max.)	Preheat temp.	100°C Max. (60 sec Max.)
Soldering time	3 sec Max.	Bath temp. & time	260°C Max., 5 sec Max.
Distance	3mm Min. (From solder joint to epoxy bulb)	Distance	2mm Min. (From solder joint to epoxy bulb)

The wave peak welding curve is recommended



- When soldering, do not put stress on the LEDs during heating.
- Dip and hand soldering should not be done more than one time.



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